# Empirical Proof Record: VALIDATED

2026-05-23T23:13:23.435527+00:00

## Empirical Proof Record — VALIDATED

**Proof ID:** `4abe91b30cd0ec9ede6b8e46d795e0a2…`
**Created:** 2026-05-23T23:13:23.435527+00:00

### 1. Claim

> Across 30 synthetic two-sample pairs of size 50 per group with varying effect sizes and variance ratios (seed=20260518), "scipy.stats.ttest_ind(equal_var=False)", "statsmodels.stats.weightstats.ttest_ind(usevar='unequal')", and "pingouin.ttest(correction=True)" produce Welch t-statistics AND two-sided p-values that agree to ≤ 1e-09 across all pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation; three-way variant with genuinely-independent statsmodels.)

- **Operationalisation:** Generate N (x, y) pairs where x ~ Normal(0, $\sigma$__x$^2$) and y ~ Normal($\delta$, $\sigma$__y$^2$) for a sweep of $\delta$ across [-1.0, 1.0] and variance-ratio $\sigma$__y/$\sigma$__x across [0.5, 2.0]. Compute Welch's t and two-sided p under each backend. Compute max_abs_diff = max over (pair, statistic ∈ {t, p}, backend-pair) of |stat_a - stat_b|. VALIDATED iff max_abs_diff ≤ tolerance.
- **Threshold:** `max_absolute_welch_difference <= 1e-09 t_or_p`
- **H0:** At least one backend pair disagrees on the Welch t-statistic OR the two-sided p-value by more than the floating-point tolerance.
- **H1:** All three backends compute identical Welch t AND p to floating-point tolerance across every pair.

### 2. Verdict

**VALIDATED** — observed `1.7763568394002505e-15`
30 pairs × 3 backends × 2 statistics; max pairwise |Δ| = 1.776e-15 on (t, scipy_vs_statsmodels) (≤ 1e-09 tol)

## 3. Evidence

| Pillar | Statistic | |
|---|---|---|
| welch_t_scipy_vs_statsmodels_vs_pingouin | max_absolute_welch_difference | 1.776356839400250 |

## 4. Pre-registration

- Registered at: 2026-05-23T23:13:23.178033+00:00
- Config hash: 650f785bb6c4c650aa559236e9e29db3…
- Data hash: 650f785bb6c4c650aa559236e9e29db3…

## 5. Signature

2b8537e99eddb60491abbe2cf7dfb828f25ba4d7571a85000a9d41d2af28a3a2